perplexity

Principle (Simple Version)

Hardness in water comes from calcium (Ca^{2+}) and magnesium (Mg^{2+}) salts.

- **Temporary hardness** is caused by bicarbonates ($Ca(HCO_3)_2$, $Mg(HCO_3)_2$) and can be removed by boiling.
- **Permanent hardness** is caused by chlorides and sulphates, which boiling cannot remove.
- **Total hardness** = temporary hardness + permanent hardness.

This experiment uses **complexometric titration** with **EDTA** and **Erichrome Black-T** indicator:

- EDTA binds Ca^{2+} and Mg^{2+} ions, removing them from the indicator-metal complex.
- At the start, the solution is wine red; when all metal ions are bound to EDTA, it turns blue.

By titrating before and after boiling, we measure total and permanent hardness, then subtract to find temporary hardness.

Procedure (Easy Steps)

1. **Estimate Total Hardness:**
    - Pipette a measured volume (V_1 mL) of the water sample into a conical flask.
    - Add ammonia buffer until you smell ammonia.
    - Add 2 drops of Erichrome Black-T indicator (solution turns wine red).
    - Titrate with EDTA from the burette until the color changes to blue.
    - Record the EDTA volume (V_2 mL) and repeat until results match (concordant).
2. **Estimate Permanent Hardness:**
    - Boil the water sample for 30 minutes to remove temporary hardness.
    - Pipette a measured volume (V_3 mL) of this boiled sample.
    - Add ammonia buffer and 2 drops of Erichrome Black-T indicator.
    - Titrate with EDTA until wine red turns blue.
    - Record the EDTA volume (V_4 mL) and repeat for concordant readings.
3. **Calculate Temporary Hardness:**
    - Temporary hardness = (Total hardness) − (Permanent hardness), based on the EDTA volumes.

Would you like me to also convert all three simplified principles and procedures into **short, bullet-style lab cheat notes** so they’re easy to recall in your exam?